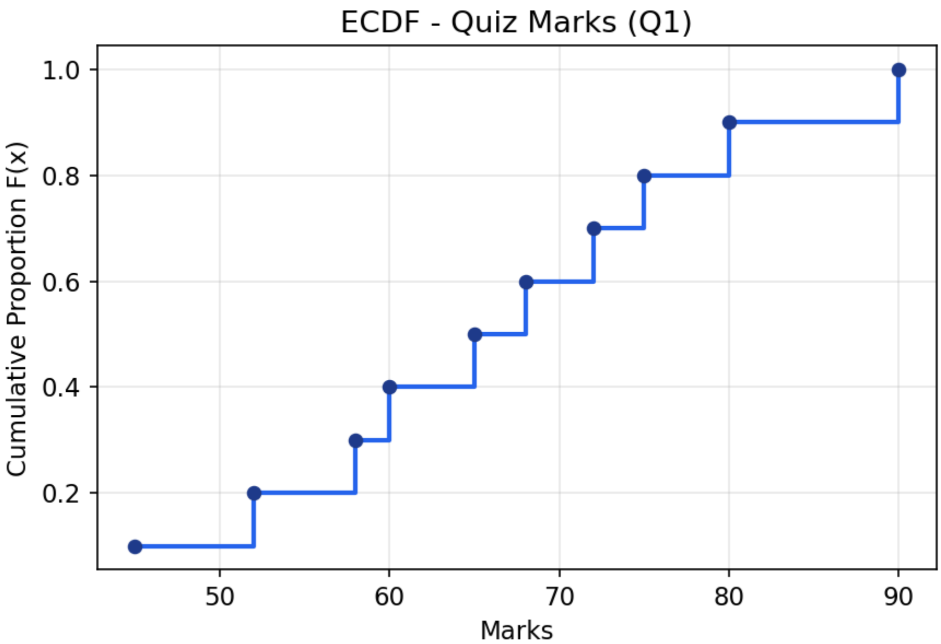
Activity 4 — Solutions

ECDF, Histogram, Density Plot, and Q-Q Plot Analysis

Question 1 — ECDF of Quiz Marks

The Empirical Cumulative Distribution Function (ECDF) estimates the proportion of observations less than or equal to each value x, using F(x) = (number of observations ≤ x) / n. With n = 10 students, each observation contributes 1/10 = 0.10 to the cumulative proportion once the data is sorted in ascending order.

Mark (x)	Rank	Cumulative Count	F(x) = Cumulative Proportion
45	1	1	0.10
52	2	2	0.20
58	3	3	0.30
60	4	4	0.40
65	5	5	0.50
68	6	6	0.60
72	7	7	0.70
75	8	8	0.80
80	9	9	0.90
90	10	10	1.00

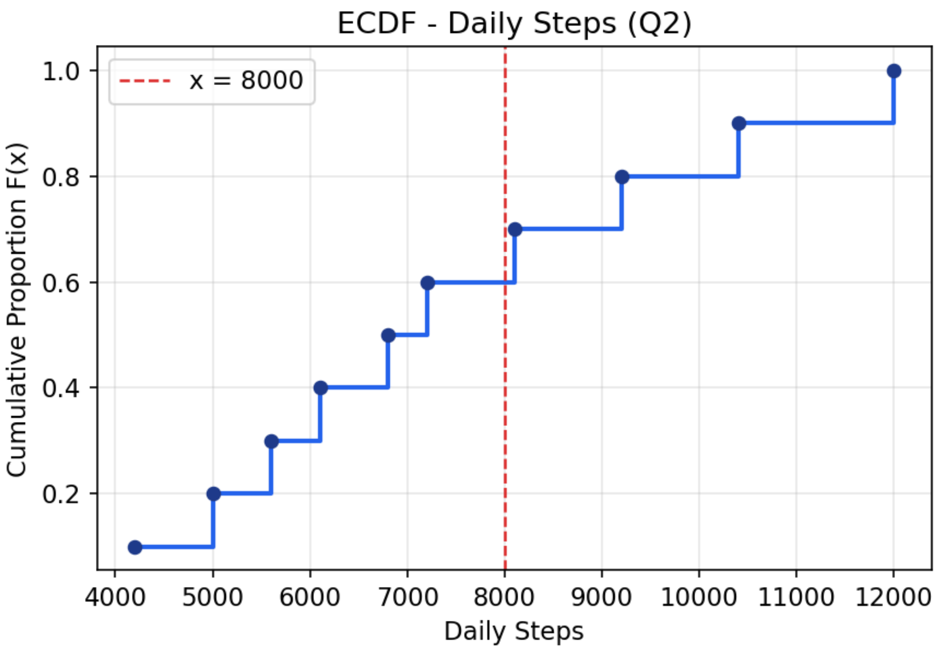


Interpretation: The ECDF rises in equal steps of 0.10 because each of the 10 marks is distinct and equally weighted. The plot shows, for example, that F(65) = 0.50 — half the class scored 65 or below. The steady, roughly linear rise (no long flat regions or steep jumps) indicates the marks are fairly evenly spread across the range 45–90, with no strong clustering at any particular score.

Question 2 — ECDF of Daily Steps

As in Question 1, the ECDF is built by sorting the 10 participants' step counts and assigning cumulative proportion i/n to the i-th smallest value.

Daily Steps (x)	Rank	Cumulative Count	F(x)
4200	1	1	0.10
5000	2	2	0.20
5600	3	3	0.30
6100	4	4	0.40
6800	5	5	0.50
7200	6	6	0.60
8100	7	7	0.70
9200	8	8	0.80
10400	9	9	0.90
12000	10	10	1.00



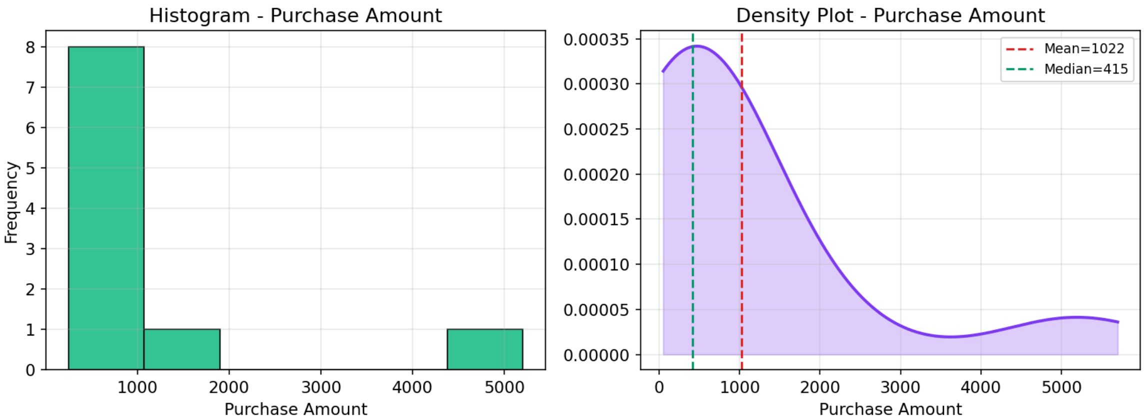
Proportion walking ≤ 8,000 steps: Six of the ten participants (P1–P6, with steps up to 7,200) walked 8,000 steps or fewer, plus P7 at exactly 8,100 does **not** qualify. So participants with steps ≤ 8000 are P1–P6 = 6 participants ⇒ F(8000) = 6/10 = **0.60 (60%)**.

Significance: The ECDF lets the trainer instantly read off what fraction of the group met or missed a target threshold (e.g., the common 8,000–10,000 steps/day goal) without needing a probability model. Here, 60% of participants fell short of 8,000 steps, highlighting that the majority of the group has room to increase activity, while the top performers (P8–P10) are well above typical daily targets.

Question 3 — Skewness of Purchase Amounts

Descriptive statistics for the purchase amounts (■) are summarized below:

Statistic	Value
Mean	1022.00
Median	415.00
Standard Deviation	1536.68
Pearson Skewness 3(Mean−Median)/SD	1.19
Fisher–Pearson Skewness (moment-based)	2.30

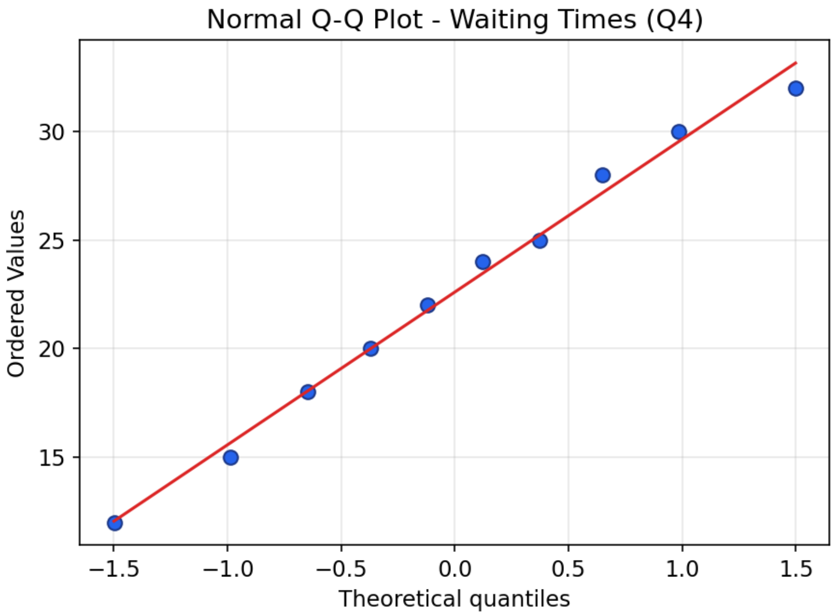


Conclusion: The distribution is **positively (right) skewed**. Justification: (1) the mean (1022) is much larger than the median (415), which happens when a few large outlying values pull the mean upward; (2) both skewness coefficients are positive and well above 0 (Pearson = 1.19, Fisher–Pearson = 2.30), confirming a right-skewed shape; and (3) the histogram shows most purchases clustered between ■250–■650, with a long tail stretching toward the two high values (■1800 and ■5200) contributed by C9 and C10. The density plot mirrors this: a tall peak near the low end and a thin tail extending right — the classic signature of positive skew.

Question 4 — Q-Q Plot of Waiting Times

A Q-Q (Quantile-Quantile) plot compares the sample quantiles of the waiting-time data against the theoretical quantiles of a Normal distribution. If the data is normally distributed, the points should fall approximately along the reference (45°) line.

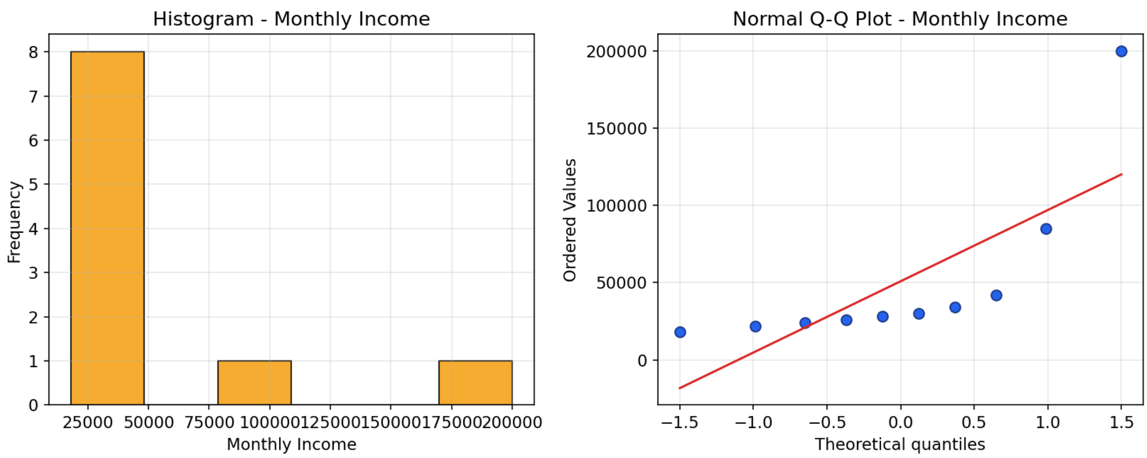
Statistic	Value
Mean	22.60 min
Median	23.00 min
Standard Deviation	6.48 min
Shapiro–Wilk statistic (W)	0.979
Shapiro–Wilk p-value	0.959



Conclusion: The waiting times **approximately follow a Normal distribution**. In the Q-Q plot, all ten points lie close to the red reference line with no systematic curvature or heavy-tail departure at either end. This is confirmed by the Shapiro–Wilk test, where the high p-value (0.959, far above the usual 0.05 threshold) means we fail to reject the null hypothesis of normality. The mean (22.6) and median (23.0) are also very close, consistent with a roughly symmetric distribution.

Question 5 — Monthly Income: Histogram & Q-Q Plot

Statistic	Value
Mean	■ 50,900.00
Median	■ 29,000.00
Standard Deviation	■ 55,758.31
Fisher–Pearson Skewness	2.18
Shapiro–Wilk statistic (W)	0.605
Shapiro–Wilk p-value	0.00006



1. **Histogram:** Nearly all customers (C1–C8) earn between ■18,000 and ■42,000, forming a tall bar at the low end, while C9 (■85,000) and C10 (■200,000) form an isolated, sparse tail far to the right.

2. **Normal Q-Q Plot:** The plot shows the eight lower-income points curving below the reference line, while the two highest incomes shoot sharply upward, well off the line.

3. **Conclusion:** The income data is **highly (positively) skewed, not normally distributed**. The mean (■50,900) is far greater than the median (■29,000); the skewness coefficient (2.18) is large and positive; and the Shapiro–Wilk test gives a p-value of 0.00006, far below 0.05, so normality is decisively rejected.

4. **How the Q-Q plot supports this:** Under normality, points should hug the diagonal reference line throughout. Here, the two extreme incomes (C9, C10) deviate sharply upward at the right end — a classic signature of a heavy right tail caused by outliers — while the bulk of points at the lower end are compressed and bend below the line. This systematic curvature (rather than a straight-line fit) is direct visual evidence of positive skew, consistent with the histogram and the numerical skewness/normality test results above.

All calculations performed using Python (NumPy, SciPy, Matplotlib): ECDF via sorted-rank cumulative proportions, skewness via Pearson's and Fisher–Pearson coefficients, and normality via the Shapiro–Wilk test.